Enabling Interoperability Between PVLens and SIDER: A Pipeline for Generating SIDER-Compatible Drug–Adverse Event Data

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INTRODUCTION & BACKGROUND

Public pharmacovigilance resources such as the SIDER database are widely used for research on drug–adverse event associations. However, SIDER has not been updated since 2015, limiting its usefulness for current safety analyses.

Newer resources such as PVLens provide more current drug labeling information but use different data structures, creating barriers for researchers who rely on existing SIDER-based analysis pipelines.

This study presents a pipeline that transforms PVLens pharmacovigilance data into a SIDER-compatible format, enabling researchers to use updated safety data without modifying existing workflows.

OBJECTIVES

- Develop a pipeline to transform PVLens drug–adverse event data into a SIDER-compatible structure.
- Compare ATC drug classification coverage between PVLens and SIDER.
- Identify overlapping and unique drug classifications between the two resources.

DATA SOURCES

Two pharmacovigilance resources were used:

PVLens database

An open-source research database derived from FDA Structured Product Labels that captures drug–adverse event information from product labeling.

SIDER (Version 4.1, 2016)

A widely used public database of drug side effects extracted from drug labeling.

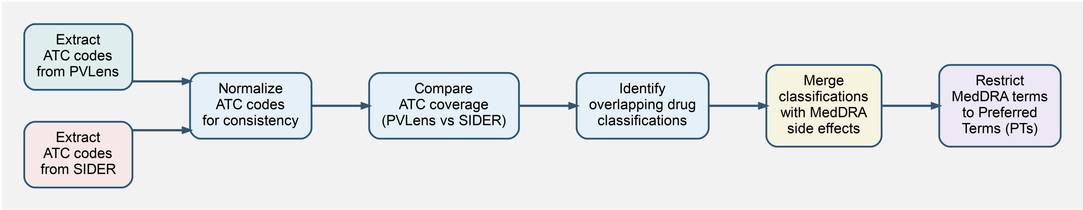

Figure 1 – Data Integration Pipeline

METHODS

A data integration pipeline (Figure 1) was developed to align drug classifications and adverse event terminology across the two datasets.

Pipeline processing steps:

- Extract ATC drug classifications from PVLens.
- Extract ATC classifications from SIDER.
- Normalize ATC codes to ensure consistent formatting.
- Compare ATC coverage across the two resources.
- Identify overlapping drug classifications.
- Generate a SIDER-compatible dataset by linking PVLens drug records to MedDRA adverse events.
- Standardize adverse event terminology at the MedDRA Preferred Term (PT) level.

The resulting dataset preserves the file structure used by SIDER-based analysis pipelines while incorporating updated PVLens data.

RESULTS

Cross-referencing the datasets revealed 1,513 overlapping ATC records and 5,385 additional ATC codes unique to PVLens (Figure 2). Only 48 SIDER ATC codes were absent from PVLens; analysis confirmed these compounds are predominantly withdrawn, discontinued, or unapproved medications. Their exclusion validates the extraction pipeline and indicates that PVLens primarily reflects currently marketed therapeutics.

The final dataset contained 190,116 drug–adverse event records standardized at the MedDRA Preferred Term level.

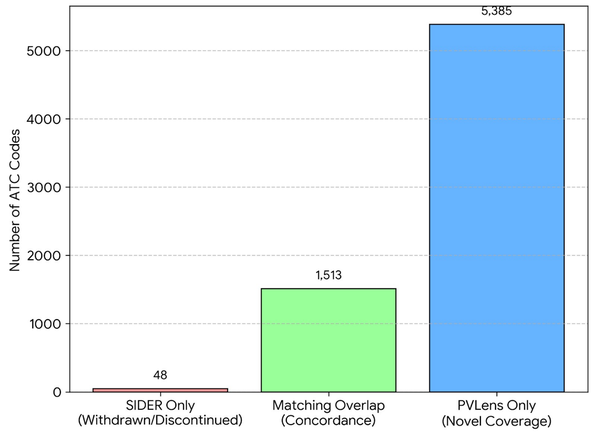

Figure 2 – ATC Coverage Comparison

DISCUSSION

The results demonstrate that PVLens substantially expands drug coverage relative to SIDER while maintaining strong overlap with drug classifications used in prior pharmacovigilance studies.

Providing PVLens data in a SIDER-compatible format enables researchers to integrate newer pharmacovigilance evidence into established workflows without modifying existing analytic pipelines.

However, several limitations remain. Because SIDER is a legacy resource, it provides only a partial benchmark for validation. Restricting adverse events to MedDRA Preferred Terms simplifies analysis but removes some granularity present in Lower Level Terms.

CONCLUSION

This study presents a pipeline for transforming PVLens pharmacovigilance data into a **SIDER-compatible format**, enabling researchers to incorporate updated drug safety information into existing SIDER workflows.

By lowering the technical barrier to adoption, this approach supports broader use of current pharmacovigilance data for drug safety research.

REFERENCES

1. Kuhn, M., Letunic, I., Jensen, L. J., & Bork, P. (2016). The SIDER database of drugs and side effects. Nucleic acids research, 44(D1), D1075-D1079.

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Source Code repository:

https://github.com/jlpainter/AMIA2026